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Battery pack

Abstract

A battery pack is provided. The battery pack includes a battery cell including an electrode extending in a first direction, a circuit portion connected to the electrode of the battery cell, and a connection portion forming a connection between the electrode of the battery cell and the circuit portion, the connection portion including a conductive thermocompression bonding layer and a conductive pad layer having a concave accommodation space formed in a second direction intersecting with the first direction to accommodate a portion of the conductive thermocompression bonding layer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6409866	12/2001	Yamada	29/829	H01L 24/32
11411256	12/2021	Lee et al.	N/A	N/A
2009/0280401	12/2008	Kim	429/120	H01M 50/581
2012/0052366	12/2011	Park	N/A	N/A
2014/0120401	12/2013	Shin et al.	N/A	N/A
2015/0044527	12/2014	Nuedecker et al.	N/A	N/A
2017/0207466	12/2016	Nishimura	N/A	H01M 50/557
2017/0208694	12/2016	Hirata et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2004172295	12/2003	JP	H05K 3/46
2009-176792	12/2008	JP	N/A
10-1156266	12/2011	KR	N/A
10-2015-0100148	12/2014	KR	N/A
10-2016-0030278	12/2015	KR	N/A
10-1956932	12/2018	KR	N/A
10-2019-0065750	12/2018	KR	N/A

OTHER PUBLICATIONS

Translation of Japanese patent ‘Circuit Board and Battery Unit’ (Year: 2004). cited by examiner
Office Action dated Dec. 29, 2023, of the corresponding CN Patent Application No. 202210256894.5. cited by applicant
European Search Report dated Oct. 31, 2022. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0034239, filed on Mar. 16, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) Embodiments relate to a battery pack.

2. Description of the Related Art

(3) In general, secondary batteries are batteries that can be charged and discharged, unlike primary batteries that cannot be charged. Secondary batteries are used as energy sources for mobile devices, electric vehicles, hybrid vehicles, electric bicycles, uninterruptible power supply, etc. Secondary batteries may be used in the form of a single battery or in the form of a module bundled as a unit by connecting a plurality of batteries, depending on the type of external device to be applied.

SUMMARY

(4) According to one or more embodiments, a battery pack includes: a battery cell including an electrode extending in a first direction; a circuit portion connected to the electrode of the battery cell; and a connection portion forming a connection between the electrode of the battery cell and the circuit portion, the connection portion including a conductive thermocompression bonding layer and a conductive pad layer having a concave accommodation space formed in a second direction intersecting with the first direction to accommodate a portion of the conductive thermocompression bonding layer.

(5) The electrode may include a first electrode and a second electrode, arranged in a third direction intersecting with the first and second directions, the first and second electrodes having different polarities, wherein the conductive pad layer may include a first conductive pad layer and a second conductive pad layer, which are apart from each other in the third direction so as to be connected to the first electrode and the second electrode, respectively.

(6) The conductive thermocompression bonding layer may be continuously formed in the third direction between the first electrode and the first conductive pad layer and between the second electrode and the second conductive pad layer.

(7) The conductive thermocompression bonding layer may be between the electrode of the battery cell and the conductive pad layer in the second direction.

(8) The conductive pad layer may include a convex portion relatively protruding in the second direction and a concave portion relatively concave in the second direction.

(9) The concave portion may provide the concave accommodation space.

(10) The conductive thermocompression bonding layer may include conductive particles supported on the convex portion and an insulating resin accommodated in the concave portion.

(11) The convex portion and the concave portion may extend in parallel in the first direction.

(12) The convex portion and the concave portion include the same metal material.

(13) The conductive pad layer may include an uneven pattern in which the convex portion and the concave portion are alternately arranged in a third direction intersecting with the first and second directions.

(14) The uneven pattern may be formed at a position biased close to a front position of the conductive pad layer facing the battery cell.

(15) The uneven pattern may be completely surrounded by an edge portion of the conductive pad layer.

(16) An insulating layer may be formed around the conductive pad layer.

(17) At least some of edge portions of the conductive pad layer forming a boundary between the conductive pad layer and the insulating layer may form a step so that an upper surface of the conductive pad layer protrudes more prominently than an upper surface of the insulating layer in the second direction.
(18) The step may be formed around an uneven pattern formed on the conductive pad layer.
(19) The step may be formed around the conductive thermocompression bonding layer.
(20) The edge portions of the conductive pad layer may include: a front edge portion formed at a position relatively close to the battery cell in the first direction; a rear edge portion formed at a position relatively far from the battery cell in the first direction; and a side edge portion connecting the front edge portion to the rear edge portion.
(21) The step may be formed in a front portion of the side edge portion overlapping the uneven pattern in a third direction intersecting with the first and second directions.
(22) The step may be formed in a front portion of the side edge portion overlapping the conductive thermocompression bonding layer in a third direction intersecting with the first and second directions.
(23) The step may be formed in the front edge portion relatively close to the battery cell in the first direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Features will become apparent to those of skill in the art by describing in detail example embodiments with reference to the attached drawings in which:
(2) FIG. **1** is a perspective view of a battery pack according to an example embodiment;
(3) FIG. **2** is an exploded perspective view of the battery pack shown in FIG. **1**;
(4) FIG. **3** is a perspective view of a portion of the battery pack shown in FIG. **1**;
(5) FIG. **4** is a cross-sectional view taken along line IV-IV of FIG. **3**;
(6) FIG. **5** is a perspective view of a portion of the battery pack shown in FIG. **1**;
(7) FIG. **6** is a plan view of a portion of the battery pack shown in FIG. **5**;
(8) FIG. **7** is a perspective view of a portion of the battery pack shown in FIG. **1**; and
(9) FIGS. **8**A and **8**B illustrate the structures of a comparative example and an example according to an embodiment, for measuring a connection resistance value between an electrode of a battery cell and a conductive pad layer.
DETAILED DESCRIPTION
(10) Example embodiments will now be described more fully hereinafter with reference to the accompanying drawings; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey example implementations to those skilled in the art. In the drawing figures, the dimensions of layers and regions may be exaggerated for clarity of illustration. Like reference numerals refer to like elements throughout.
(11) As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.
(12) Hereinafter, a battery pack according to an example embodiment will be described with reference to the accompanying drawings.
(13) FIG. **1** is a perspective view of a battery pack according to an example embodiment. FIG. **2** is an exploded perspective view of the battery pack shown in FIG. **1**. FIG. **3** is a perspective view of a portion of the battery pack shown in FIG. **1**. FIG. **4** is a cross-sectional view taken along line IV-IV

of FIG. **3**. FIG. **5** is a perspective view of a portion of the battery pack shown in FIG. **1**. FIG. **6** is a plan view of a portion of the battery pack shown in FIG. **5**. FIG. **7** is a perspective view of a portion of the battery pack shown in FIG. **1**.
(14) Referring to FIGS. **1** and **2**, a battery pack according to an example embodiment may include a battery cell **10**, a circuit portion C, and a connection portion **56**.
(15) The battery cell **10** may include electrodes **11** and **12** extending in a first direction Z**1**.
(16) The circuit portion C may be connected to the electrodes **11** and **12** of the battery cell **10**.
(17) The connection portion **56** may form a connection between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C. The connection portion **56** may include a conductive thermocompression bonding layer **50** and a conductive pad layer **60**. As described in additional detail below, the conductive pad layer **60** may have a concave accommodation space formed in a second direction Z**2** (which intersects with the first direction Z**1**) to accommodate a portion of the conductive thermocompression bonding layer **50**.
(18) The battery cell **10** may include an electrode assembly **10***a*, a casing **10***b* surrounding the electrode assembly **10***a*, and the electrodes **11** and **12** drawn out from the casing **10***b*.
(19) Although not shown in the drawings, the electrode assembly **10***a* may be formed as a winding type, in which first and second electrode plates and a separator between the first electrode plate and the second electrode plate are wound in a roll shape, or may be formed in a stacked type, in which the first and second electrode plates and the separator are stacked on each other.
(20) The first and second electrode plates of the electrode assembly **10***a* may be electrically connected to the outside of the casing **10***b* through the electrodes **11** and **12** of the battery cell **10**. The electrodes **11** and **12** of the battery cell **10** may be electrically connected to the first and second electrode plates of the electrode assembly **10***a*, respectively. The first electrode **11** and the second electrode **12** may have different polarities, respectively.
(21) The casing **10***b* may be formed to surround the electrode assembly **10***a*. By sealing an excess portion of the casing **10***b* remaining after surrounding the electrode assembly **10***a*, a sealing portion TS for sealing the electrode assembly **10***a* may be formed. In this case, the battery cell **10** may include a body **10***c* including the electrode assembly **10***a* and the casing **10***b* surrounding the electrode assembly **10***a*, and a sealing portion TS formed along the periphery of the body **10***c* and including the casing **10***b* for sealing the electrode assembly **10***a*. In this case, the sealing portion TS may include the terrace portion T from which the electrodes **11** and **12** are drawn out, and may include, in addition to the terrace portion T, a side sealing portion S formed along the side of the body **10***c* of the battery cell **10**.
(22) The electrodes **11** and **12** of the battery cell **10** may be drawn out through the terrace portion T, and may be electrically connected to the circuit portion C through the connection portion **56**, as will be described below. In an example embodiment, the electrodes **11** and **12** of the battery cell **10** may extend in the first direction Z**1**, and may be electrically connected to the circuit portion C arranged at a front position of the battery cell **10** in the first direction Z**1**.
(23) In an example embodiment, the first direction Z**1** may mean a direction in which the electrodes **11** and **12** of the battery cell **10** extend, or may mean a direction in which the battery cell **10** and the circuit portion C are arranged, or a direction in which the battery cell **10** and the circuit portion C face each other. In this case, the front position of the battery cell **10** in the first direction Z**1** may mean a front position of the battery cell facing the circuit portion C. Similarly, the front position of the circuit portion C in the first direction Z**1** may mean a front position of the circuit portion C facing the battery cell **10**.
(24) The electrodes **11** and **12** of the battery cell **10** may include the first electrode **11** and the second electrode **12**, arranged in a third direction Z**3**. As will be described below, the first and second electrodes **11** and **12** having different polarities may be respectively connected to first and second conductive pad layers **61** and **62** arranged on the circuit portion C. A detailed technical configuration of the conductive pad layer **60** including the first and second conductive pad layers

61 and **62** will be described below in more detail.
(25) The circuit portion C may form a charge and discharge path of the battery cell **10**. In an example embodiment, the circuit portion C is to form a charge and discharge path connected to the electrodes **11** and **12** of the battery cell **10**, and may form a charge and discharge path between the battery cell **10** and an external device. In an example embodiment, the external device may correspond to an external load receiving discharge power from the battery cell **10** or an external charger providing charging power toward the battery cell **10**, and the circuit portion C may form a discharge path from the battery cell **10** to an external load or form a charge path from an external charger to the battery cell **10**. In various embodiments, the circuit portion C may include the entire charge and discharge path between the battery cell **10** and an external device, or may include only a portion of the charge and discharge path between the battery cell **10** and the external device.
(26) When the battery pack according to an example embodiment includes the circuit portion C electrically connected to the battery cell **10**, the circuit portion C may include the entire charge and discharge path between the battery cell **10** and an external device, or may include only a portion of the charge and discharge path between the battery cell **10** and the external device. For example, the battery pack according to an example embodiment may include, as the circuit portion C electrically connected to the battery cell **10**, various components as long as the circuit portion C is electrically connected to the battery cell **10** to form a charge and discharge path of the battery cell **10**, and may not necessarily include all components connected to an external device.
(27) The circuit portion C according to an example embodiment may include a circuit board connected to the battery cell **10**, i.e., a circuit board having a conductive line for forming a charge and discharge path of the battery cell **10**, such as a rigid circuit board including a relatively rigid insulating substrate or a flexible insulating board including a relatively soft insulating film. On the circuit board, a circuit element (not shown), which is capable of obtaining state information such as the voltage, current, and temperature of the battery cell **10** or is capable of controlling the charging and discharging operations of the battery cell **10** based on the obtained state information of the battery cell **10**, may be arranged. In the drawings, the circuit portion C may include a circuit board including an insulating layer **80**, and a conductive line (not shown) for forming a charge and discharge path of the battery cell **10** may be formed on the circuit board. In an example embodiment, the conductive line (not shown) may extend from the connection portion **56** connected to the electrodes **11** and **12** of the battery cell **10**. When it is assumed throughout the present specification that the connection portion **56** forms a conductive connection between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C, it may mean that the connection portion **56** forms a conductive connection between the electrodes **11** and **12** of the battery cell **10** and a conductive line of the circuit board, e.g., a conductive line formed on the insulating layer **80**.
(28) In an example embodiment, the circuit portion C may include a safety element (not shown) prepared on the charge and discharge path of the battery cell **10** and capable of blocking charge and discharge current of the battery cell **10** by capturing abnormal conditions such as overheating, overcharging, and overdischarging of the battery cell **10**. For example, in an example embodiment, in order to forcibly reduce or block the charge and discharge current according to the overheating of the battery cell **10** when the battery cell **10** is overheated above a preset threshold temperature, the safety element (not shown) may be prepared on the charge and discharge path of the battery cell **10** and may include a variable resistor having resistance that varies with temperature. For example, in an example embodiment, the safety element may include a positive temperature coefficient (PTC), a thermal cut-off (TCO), or the like. For example, the circuit portion C may include both a circuit board on which a conductive line for forming a charge and discharge path of the battery cell **10** is formed and a safety element (not shown) arranged on the circuit board, or may include only a safety element (not shown) that is not supported by the circuit board.
(29) In an example embodiment, the connection portion **56** may be between the battery cell **10** and

the circuit portion C. The connection portion **56** may be between the battery cell **10** and the circuit portion C to form an electrical connection therebetween.
(30) In an example embodiment, the connection portion **56** may include a conductive thermocompression bonding layer **50** and a conductive pad layer **60** having a concave accommodation space formed in the second direction Z**2** to accommodate a portion of the conductive thermocompression bonding layer **50**. In an example embodiment, the second direction Z**2** may refer to a thickness direction of the conductive thermocompression bonding layer **50** or a thickness direction of the conductive pad layer **60**, and may refer to a direction in which the conductive thermocompression bonding layer **50** is compressed or a direction in which the conductive pad layer **60** is compressed. The second direction Z**2** may correspond to a direction intersecting with the first direction Z**1** in which the electrodes **11** and **12** of the battery cell **10** extend and the third direction Z**3** in which the electrodes **11** and **12** of the battery cell **10** are arranged. In an example embodiment, the second direction Z**2** may correspond to a direction perpendicular to the first and third directions Z**1** and Z**3**.
(31) Referring to FIGS. **3** and **4**, the conductive thermocompression bonding layer **50**, which forms an electrical connection between the battery cell **10** and the circuit portion C, may have conductivity through thermocompression bonding between the battery cell **10** and the circuit portion C. In an example embodiment, the conductive thermocompression bonding layer **50** may refer to a component that does not have conductivity before thermocompression bonding, but has conductivity through thermocompression bonding. The conductive thermocompression bonding layer **50** according to an example embodiment is different from a metal member that may be recognized as a conductive material regardless of thermocompression bonding, and may refer to a component in which conductivity thereof may be differentiated before thermocompression bonding and after thermocompression bonding. That is, in an example embodiment, the conductive thermocompression bonding layer **50** may form an electrical connection between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C through the conductive transition of the conductive thermocompression bonding layer **50**, i.e., a transition from a non-conductive state (or an insulating state) before thermocompression bonding to a conductive state after thermocompression bonding. Throughout the present specification, the conductive transition of the conductive thermocompression bonding layer **50** may refer to a transition from a non-conductive state (or an insulating state) before thermocompression bonding to a conductive state after thermocompression bonding, and may refer to a transition to a conductive state through thermocompression bonding.
(32) Referring to FIG. **4**, in an example embodiment, the conductive thermocompression bonding layer **50** may include conductive particles **51** and an insulating resin **52** accommodating the conductive particles **51**.
(33) In an example embodiment, the insulating resin **52** may be in a solid phase below a transition temperature to thereby fix the conductive particles **51**, and above the transition temperature, the insulating resin **52** may be changed to a liquid or gel phase that may have fluidity other than a solid phase and thus the conductive particles **51** dispersed in the insulating resin **52** may have fluidity. The conductive particles **51** having fluidity may be arranged between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C to form a conductive connection. For example, as the insulating resin **52**, which has fluidity above the transition temperature, is pushed out between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C by a pressure provided with heat, i.e., a pressure provided in the second direction Z**2** to make the electrodes **11** and **12** of the battery cell **10** and the circuit portion C approach each other, the remaining conductive particles **51** may electrically connect the electrodes **11** and **12** of the battery cell **10** to the circuit portion C.
(34) The conductive pad layer **60** may be formed on the circuit portion C facing the electrodes **11** and **12** of the battery cell **10**. Together with the conductive thermocompression bonding layer **50**, the conductive pad layer **60** may form the connection portion **56** electrically connecting the battery

cell **10** to the circuit portion C. In this case, the conductive pad layer **60** may help a conductive transition of the conductive thermocompression bonding layer **50** so that the conductive transition of the conductive thermocompression bonding layer **50** may be smoothly performed.

(35) Hereinafter, a configuration will be described in which the conductive thermocompression bonding layer **50** and the conductive pad layer **60**, which form the connection portion **56**, cooperate with each other to form a conductive connection between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C according to thermocompression bonding.

(36) The conductive pad layer **60** may include a convex portion **65***a* that is relatively protruded in the second direction Z**2**, and a concave portion **65***b* that is relatively concave in the second direction Z**2**. In this case, the second direction Z**2** may correspond to a thickness direction of the conductive pad layer **60**, and may correspond to a direction in which the conductive thermocompression bonding layer **50** is compressed, as will be described below. The conductive pad layer **60** may have a configuration that may help the conductive transition of the conductive thermocompression bonding layer **50**. For example, the convex portion **65***a* may provide a support base that supports the conductive particles **51** of the conductive thermocompression bonding layer **50**, and the concave portion **65***b* may provide a concave accommodation space capable of accommodating the insulating resin **52** of the conductive thermocompression bonding layer **50**.

(37) In an example embodiment, the convex portion **65***a* and the concave portion **65***b* of the conductive pad layer **60** may form an uneven pattern **65** while being alternately arranged with respect to each other. For example, the convex portion **65***a* and the concave portion **65***b* of the conductive pad layer **60** may be alternately arranged in the third direction Z**3** intersecting with the first direction Z**1** in which the electrodes **11** and **12** of the battery cell **10** extend.

(38) The conductive pad layer **60** may be formed by, e.g., selectively etching a plating layer formed on the circuit portion C. For example, a conductive pad layer **60** may be formed in which a convex portion **65***a* (which is not etched in a depth direction (i.e., the second direction Z**2**) of the plating layer) and a concave portion **65***b* (which is etched in the depth direction) are alternately arranged. In an example embodiment, the convex portion **65***a* and the concave portion **65***b*, which form the conductive pad layer **60**, may be formed by selectively etching a plating layer including a metal material, and may include the same metal material. For example, the convex portion **65***a* and the concave portion **65***b* may include the same metal material to have substantially the same conductivity.

(39) The convex portion **65***a* and the concave portion **65***b* may be alternately arranged in the third direction Z**3** to form the uneven pattern **65**. The uneven pattern **65** may include a plurality of convex portions **65***a* and a plurality of concave portions **65***b*, which are arranged in the third direction Z**3**. In an example embodiment, the convex portion **65***a* may include a plurality of convex portions **65***a* spaced apart from each other with the concave portion **65***b* therebetween. Each of the convex portions **65***a* and the concave portions **65***b*, arranged in plurality in the third direction Z**3**, may extend in the first direction Z**1**, and the convex portions **65***a* and the concave portions **65***b* may extend in parallel in the first direction Z**1**.

(40) Referring to FIGS. **5** to **7**, the conductive pad layer **60** may include a base portion **60***a* surrounding the uneven pattern **65** in which the plurality of convex portions **65***a* and the plurality of concave portions **65***b* are formed.

(41) The base portion **60***a* may form edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60**. For example, the base portion **60***a* may form the front edge portion **60**F at a front position relatively close to the battery cell **10** so as to face the battery cell **10** in the first direction Z**1**, the rear edge portion **60**R at a rear position relatively far from the battery cell **10**, and the side edge portion **60**S connecting the front edge portion **60**F to the rear edge portion **60**R. In an example embodiment, the conductive pad layer **60** may include a first conductive pad layer **61** and a second conductive pad layer **62**, which are apart from each other. The edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60** may include edge portions **60**F, **60**S, and **60**R of each of the first and

second conductive pad layers **61** and **62**. The side edge portion **60**S may include a side edge portion **60**S of each of the first and second conductive pad layers **61** and **62**. The base portion **60***a* may form the edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60**, may entirely surround the uneven pattern **65** formed in an inner region of the conductive pad layer **60**, and may separately surround the uneven pattern **65** of each of the first and second conductive pad layers **61** and **62**.
(42) In an example embodiment, the uneven pattern **65** may correspond to a portion of the conductive pad layer **60** that forms physical contact with the electrodes **11** and **12** of the battery cell **10**. The uneven pattern **65** may be formed at a position biased toward a front position of the conductive pad layer **60** relatively close to the battery cell **10** in the first direction Z**1**. Thus, the rear edge portion **60**R (extending long in the first direction Z**1** and having a flat plate shape) may be formed at a rear position of the conductive pad layer **60** that is relatively far from the battery cell **10**.
(43) As will be described below, around the uneven pattern **65** and around the conductive thermocompression bonding layer **50**, a step P may be formed so that the upper surface of the conductive pad layer **60** protrudes more prominently than the upper surface of the insulating layer **80** surrounding the conductive pad layer **60**. The step P may be formed in the front edge portion **60**F facing the battery cell **10** and a front portion of the side edge portion **60**S overlapping the uneven pattern **65** or the conductive thermocompression bonding layer **50** in the third direction Z**3**. Technical details regarding the step P between the conductive pad layer **60** and the insulating layer **80** will be described below in more detail.
(44) The base portion **60***a* may protect the uneven pattern **65** while surrounding the uneven pattern **65**, and may support the uneven pattern **65**. For example, the base portion **60***a* may be formed at the same level as the convex portion **65***a* of the uneven pattern **65** in the second direction Z**2**, and may support a plurality of convex portions **65***a* while extending across one end and the other end of each of the plurality of convex portions **65***a* in the third direction Z**3**. The base portion **60***a* or the edge portions **60**F, **60**S, and **60**R may entirely surround the uneven pattern **65** while extending in the first and third directions Z**1** and Z**3** to surround the uneven pattern **65**.
(45) In an example embodiment, the conductive pad layer **60** may be formed by selectively etching a plating layer formed on the circuit portion C. In this case, a non-etched portion may form the base portion **60***a* and the convex portion **65***a*, and a selectively etched portion may form the concave portion **65***b*. In an example embodiment, the conductive pad layer **60** may include a same metal material, as a whole. Thus, the base portion **60***a*, the convex portion **65***a*, and the concave portion **65***b*, which form the conductive pad layer **60**, may all include the same metal material.
(46) In an example embodiment, the metal material may be a metal material having excellent conductivity, and may include, e.g., aluminum or nickel.
(47) In an example embodiment, the conductive pad layer **60** may include two or more different metal materials, e.g., first and second metal materials stacked on each other. Even in this case, because the base portion **60***a*, the convex portion **65***a*, and the concave portion **65***b*, which form the conductive pad layer **60**, all include first and second metal materials stacked on each other, it may be said that the base portion **60***a*, the convex portion **65***a*, and the concave portion **65***b* include the same metal material.
(48) As will be described below, the convex portion **65***a* may be connected to the electrodes **11** and **12** of the battery cell **10** through the conductive particles **51** (refer to FIG. **4**) of the conductive thermocompression bonding layer **50**. In addition, the convex portion **65***a* and the base portion **60***a* connected to the convex portion **65***a* may be connected to the circuit portion C. The conductive pad layer **60** including the convex portion **65***a* and the base portion **60***a* may form the charge and discharge path of the battery cell **10** between the battery cell **10** and the circuit portion C. Thus, the base portion **60***a* may provide a low-resistance charge and discharge path having a larger area than the convex portion **65***a*.
(49) In an example embodiment, the electrodes **11** and **12** of the battery cell **10** may be connected

to the circuit portion C through the convex portion **65***a* and the base portion **60***a*. The concave portion **65***b* including the same metal material as the convex portion **65***a* and the base portion **60***a* may also form a charge and discharge path through which the charge and discharge current of the battery cell **10** flows. However, as will be described below, the concave portion **65***b* may provide an accommodation space for accommodating the insulating resin **52** rather than being directly connected to the electrodes **11** and **12** of the battery cell **10** through the conductive particles **51** (refer to FIG. **4**).

(50) Throughout the present specification, that the conductive particles **51** (refer to FIG. **4**) of the conductive thermocompression bonding layer **50** are connected to the electrodes **11** and **12** of the battery cell **10** does not mean that the formation of the charge and discharge path of the battery cell **10** is limited to the convex portion **65***a* providing a support base for the conductive particles **51** and the base portion **60***a* connected to the convex portion **65***a*. For example, in various embodiments of the disclosure, the charge and discharge path of the battery cell **10** may be formed through the entire conductive pad layer **60** including the uneven pattern **65** and the base portion **60***a*, the uneven pattern **65** including the convex portion **65***a* and the concave portion **65***b*. Similarly, in the present specification, that the battery cell **10** and the circuit portion C are connected to each other through the conductive particles **51** (refer to FIG. **4**) arranged on the convex portion **65***a* does not mean that the charge and discharge path of the battery cell **10** is limited to the convex portion **65***a*, and the charge and discharge path of the battery cell **10** may be formed through part of the conductive pad layer **60** including the convex portion **65***a* or all of the conductive pad layer **60**.

(51) Referring to FIG. **4**, the conductive pad layer **60** may be configured such that conductive transition of the conductive thermocompression bonding layer **50** that forms the connection portion **56** together with the conductive pad layer **60** may be smoothly performed. The convex portion **65***a* forming the uneven pattern **65** of the conductive pad layer **60** may provide a support base for the conductive particles **51** of the conductive thermocompression bonding layer **50**. The concave portion **65***b* may provide a concave accommodation space capable of accommodating the insulating resin **52** of the conductive thermocompression bonding layer **50**. When the conductive thermocompression bonding layer **50** is heated above the transition temperature, the insulating resin **52** of the conductive thermocompression bonding layer **50** may have fluidity, and may be accommodated in the concave accommodation space, provided by the concave portion **65***b*, while being pushed from the convex portion **65***a* into the concave portion **65***b* according to a pressure applied between the electrodes **11** and **12** of the battery cell **10** and the circuit portion C (i.e., according to a pressure applied in the second direction Z**2** to make the electrodes **11** and **12** of the battery cell **10** and the circuit portion C approach each other), and the conductive particles **51** remaining on the convex portion **65***a* may electrically connect the electrodes **11** and **12** of the battery cell **10** to the circuit portion C.

(52) The concave portion **65***b* may provide an accommodation space for accommodating the insulating resin **52** of the conductive thermocompression bonding layer **50**, and may provide an accommodation space having sufficient volume to accommodate the insulating resin **52** so that the insulating resin **52** pushed into the accommodation space of the insulating resin **52** does not interfere with a conductive contact between the conductive particles **51** remaining on the convex portion **65***a* and the battery cell **10**.

(53) By way of comparison, a general structure may have a conductive pad layer **60** on a flat plate that is not provided with the uneven pattern **65**, while being connected to the electrodes **11** and **12** of the battery cell **10**. In such a general structure, because the conductive pad layer **60** on the flat plate does not provide a differentiated support base for each component of the conductive thermocompression bonding layer **50**, the insulating resin **52** and the conductive particles **51** (which form the conductive thermocompression bonding layer **50**) may be mixed and the insulating resin **52** may interfere with the contact of the conductive particles **51** to the electrodes **11** and **12** of the battery cell **10**. This may result in an increase in electrical resistance between the electrodes **11**

and **12** of the battery cell **10** and the conductive pad layer **60**.
(54) On the other hand, according to an example embodiment, the conductive pad layer **60** is connected to the electrodes **11** and **12** of the battery cell **10** through the uneven pattern **65** formed on the conductive pad layer **60**. Thus, a differentiated support base may be provided for each component of the conductive thermocompression bonding layer **50** (i.e., the conductive particles **51** and the insulating resin **52**) through the uneven pattern **65** of the conductive pad layer **60**. The conductive pad layer **60** may support the conductive particles **51** through the convex portions **65***a* of the uneven pattern **65**, and thus, an electrical connection between the conductive particles **51** remaining on the convex portion **65***a* and the electrodes **11** and **12** of the battery cell **10** may be made. In addition, as the insulating resin **52** is accommodated through the concave portion **65***b* of the uneven pattern **65**, the insulating resin **52** may not interfere with the contact between the conductive particles **51** remaining and placed on the convex portion **65***a* and the electrodes **11** and **12** of the battery cell **10**.
(55) Referring again to FIG. **3**, in the battery cell **10** according to an example embodiment, the first and second electrodes **11** and **12** having different polarities may be arranged in the third direction Z**3** intersecting with the first direction Z**1** in which the first and second electrodes **11** and **12** extend. The first and second electrodes **11** and **12** may be respectively connected to the first and second conductive pad layers **61** and **62**, which are provided separately. That is, in an example embodiment, the conductive pad layer **60** may include the first and second conductive pad layers **61** and **62** that are apart from each other in the third direction Z**3** and are respectively connected to the first and second electrodes **11** and **12** that are different from each other. In an example embodiment, the first and second conductive pad layers **61** and **62** may be electrically connected to the first and second electrodes **11** and **12** of the battery cell **10** through one conductive thermocompression bonding layer **50** continuously extending across the first and second conductive pad layers **61** and **62** in the third direction Z**3**.
(56) In an example embodiment, the conductive thermocompression bonding layer **50** may have conductivity in the second direction Z**2**, and may electrically connect the first and second electrodes **11** and **12** of the battery cell **10** to the first and second conductive pad layers **61** and **62**, the first and second electrodes **11** and **12** and the first and second conductive pad layers **61** and **62** being arranged to overlap each other in the second direction Z**2** with the conductive thermocompression bonding layer **50** therebetween. However, the conductive thermocompression bonding layer **50** may have no conductivity in the third direction Z**3**, and thus may not cause an electrical short circuit between the first and second electrodes **11** and **12** of the battery cell **10** or between the first and second conductive pad layers **61** and **62**. In an example embodiment, the conductive thermocompression bonding layer **50** may have anisotropy, in which conductive properties change depending on the direction, so that the conductive thermocompression bonding layer **50** has conductivity in the second direction Z**2** and non-conductivity (or insulation) in the third direction Z**3**.
(57) As described above, in an example embodiment, the first and second conductive pad layers **61** and **62** respectively connected to the first and second electrodes **11** and **12** having different polarities may be provided to be separated from each other, but the conductive thermocompression bonding layer **50** connecting the first electrode **11** to the first conductive pad layer **61** and connecting the second electrode **12** to the second conductive pad layer **62** may be provided as a single member continuously formed. Thus, the connection between the first and second electrodes **11** and **12** and the first and second conductive pad layers **61** and **62** may be collectively formed through one operation. In addition, the first and second electrodes **11** and **12** may be electrically connected to the first and second conductive pad layer **61** and **62**, respectively, in the second direction Z**2** by using the conductive thermocompression bonding layer **50** having anisotropy, which has different conductive properties depending on the direction, and an electrical short circuit between the first and second electrodes **11** and **12** and between the first and second conductive pad

layers **61** and **62** in the third direction Z**3** may also be prevented.

(58) Referring to FIGS. **3** and **4**, the conductive thermocompression bonding layer **50** according to an example embodiment may have conductivity according to thermocompression bonding. In this case, the conductive thermocompression bonding layer **50** may have conductivity according to a compression direction (corresponding to the second direction Z**2**) and may have non-conductivity (or insulation) in a direction different from the compression direction, e.g., in a direction perpendicular to the compression direction. For example, in an example embodiment, the conductive thermocompression bonding layer **50** may connect the electrodes **11** and **12** of the battery cell **10** to the conductive pad layer **60** while being compressed in the second direction Z**2** corresponding to the compression direction. The conductive thermocompression bonding layer **50** may include a plurality of conductive particles **51** dispersed on the insulating resin **52**. In addition, while the insulating resin **52** having fluidity according to heating is pushed in the second direction Z**2** corresponding to the compression direction, the remaining conductive particles **51** may connect the electrodes **11** and **12** of the battery cell **10** to the conductive pad layer **60**, the electrodes **11** and **12** and the conductive pad layer **60** overlapping each other in the second direction. However, in the third direction Z**3** that does not correspond to the compression direction, electrical connection through the conductive particles **51** may be blocked as a plurality of conductive particles **51** are discontinuously arranged or the insulating resin **52** fills gaps between the plurality of conductive particles **51**. In an example embodiment, the conductive thermocompression bonding layer **50** may include an anisotropic conductive film (ACF).

(59) Referring to FIGS. **5** to **7**, an insulating layer **80** surrounding the conductive pad layer **60** may be formed around the conductive pad layer **60**. The insulating layer **80** may provide insulation with respect to the conductive pad layer **60**. For example, the insulating layer **80** may provide electrical insulation between the conductive pad layer **60** and another conductive line of the circuit portion C.

(60) In an example embodiment, the circuit portion C may include another conductive line insulated from a conductive line conducting electricity with the conductive pad layer **60**. In this case, the insulating layer **80** may provide electrical insulation between the conductive pad layer **60** and the other conductive line.

(61) The conductive pad layer **60** may be surrounded by the insulating layer **80**. Throughout the present specification, that the conductive pad layer **60** is surrounded by the insulating layer **80** may mean that, as the conductive pad layer **60** is formed on the insulating layer **80** in the second direction Z**2**, the conductive pad layer **60** formed with a relatively narrow area is formed on the insulating layer **80** formed with a relatively large area. In an example embodiment, the conductive pad layer **60** may be formed in an inner region of the insulating layer **80** on a plane formed by the first and third directions Z**1** and Z**3**, and the thickness of the conductive pad layer **60** in the second direction Z**2** may be exposed from the insulating layer **80**. That is, throughout the present specification, that the conductive pad layer **60** is surrounded by the insulating layer **80** may mean that the conductive pad layer **60** is formed in the inner region of the insulating layer **80** on a plane formed by the first and third directions Z**1** and Z**3**. In this case, the conductive pad layer **60** may be formed in the inner region of the insulating layer **80** on a plane formed by the first and third directions Z**1** and Z**3**, and the thickness of the conductive pad layer **60** may be exposed from the insulating layer **80** in the second direction Z**2**.

(62) In an example embodiment, the thickness of the conductive pad layer **60** may be surrounded by the insulating layer **80** in the second direction Z**2**. For example, the conductive pad layer **60** may be formed in the inner region of the insulating layer **80** on a plane formed by the first and third directions Z**1** and Z**3**, and at the same time, the thickness of the conductive pad layer **60** may be surrounded by the insulating layer **80** in the second direction Z**2**. However, as will be described below, even when the thickness of the conductive pad layer **60** is surrounded by the insulating layer **80**, the thickness of the conductive pad layer **60** may be exposed around the uneven pattern **65** along the edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60**. That is, the upper

surface of the conductive pad layer **60** around the uneven pattern **65** may be formed at a level higher than the upper surface of the insulating layer **80**, and the upper surface of the conductive pad layer **60** and the upper surface of the insulating layer **80** may be formed to be stepped in the second direction Z**2**. In other words, a step P, which is formed by the upper surface of the conductive pad layer **60** and the upper surface of the insulating layer **80**, may be formed around the uneven pattern **65** along the edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60**.
(63) Referring to FIG. **3**, in an example embodiment, the step P between the conductive pad layer **60** and the insulating layer **80** may be formed around the uneven pattern **65** in a boundary between the conductive pad layer **60** and the insulating layer **80**. In this case, the periphery of the uneven pattern **65** may include a boundary, which overlaps the uneven pattern **65** in at least the third direction Z**3**, in the boundary between the conductive pad layer **60** and the insulating layer **80**. The step P between the conductive pad layer **60** and the insulating layer **80** may significantly protrude the upper surface of the conductive pad layer **60** rather than the upper surface of the insulating layer **80**, thereby providing effective pressure concentration for the conductive thermocompression bonding layer **50** formed on the conductive pad layer **60** and preventing a pressure gap of a pressing tool TO due to physical interference with the insulating layer **80**.
(64) In the thermocompression bonding, a certain heat and pressure may be provided to the conductive thermocompression bonding layer **50** between the electrodes **11** and **12** of the battery cell **10** and the conductive pad layer **60**, thereby forming a conductive connection between the electrodes **11** and **12** of the battery cell **10** and the conductive pad layer **60**. In this case, through the pressing tool TO pressing in the second direction Z to make the electrodes **11** and **12** of the battery cell **10** and the conductive pad layer **60** approach each other, the electrodes **11** and **12** and the conductive pad layer **60** being arranged to overlap each other with the conductive thermocompression bonding layer **50** therebetween, a certain heat and pressure may be applied to the conductive thermocompression bonding layer **50**. In this case, a region pressed by the pressing tool TO corresponds to a pressing region to which the pressing tool TO is projected in the second direction Z**2**. In the pressing region of the pressing tool TO, the conductive pad layer **60** to be pressed may protrude in the second direction Z**2** relative to the insulating layer **80** surrounding the conductive pad layer **60**, and thus, effective pressurization of the conductive pad layer **60** rather than the insulating layer **80** may be made, and interference or a pressure gap caused by the insulating layer **80** may be prevented.
(65) Referring to FIGS. **5** to **7**, in an example embodiment, the step P between the upper surface of the conductive pad layer **60** and the upper surface of the insulating layer **80** may be formed around the uneven pattern **65** along the edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60**, and may be formed in a side edge portion **60**S of the conductive pad layer **60**, which overlaps the uneven pattern in the third direction Z**3**. In an example embodiment, the step P of the conductive pad layer **60** may also be formed in a front edge portion **60**F, formed at a position relatively close to the battery cell **10** to face the battery cell **10** in the first direction Z**1**, as well as the side edge portion **60**S. In this case, the front edge portion **60**F of the conductive pad layer **60** may refer to an edge portion, which faces the battery cell **10**, i.e., is formed at a position relatively close to the battery cell **10**, from among the edge portions **60**F, **60**S, and **60**R of the conductive pad layer **60**. In an example embodiment, a rear edge portion **60**R (corresponding to the thickness of the rear edge portion **60**R) of the conductive pad layer **60** formed at a position relatively far from the battery cell **10** may be surrounded by the insulating layer **80** in the second direction Z**2**. For example, the insulating layer **80** may include a thick portion **80***a* formed to have a relatively large thickness to surround the rear edge portion **60**R (corresponding to the thickness of the rear edge portion **60**R).
(66) In an example embodiment, the conductive pad layer **60** may include a side edge portion **60**S connecting the front edge portion **60**F to the rear edge portion **60**R, in addition to the front edge portion **60**F and the rear edge portion **60**R. In an example embodiment, in a front portion of the front edge portion **60**F and the side edge portion **60**S each formed around the uneven pattern **65**, a

step P for protruding the upper surface of the conductive pad layer **60** may be formed, and the rear edge portion **60**R (corresponding to the thickness of the rear edge portion **60**R) far from the uneven pattern **65** and a rear portion of the side edge portion **60**S (corresponding to the thickness of the side edge portion **60**S) may be surrounded by the insulating layer **80**. For example, the insulating layer **80** may include a thick portion **80***a* formed to have a relatively large thickness to surround the rear edge portion **60**R (corresponding to the thickness of the rear edge portion **60**R) and the rear portion (corresponding to the thickness of the side edge portion **60**S) of the side edge portion **60**S. In an example embodiment, the thick portion **80***a* of the insulating layer **80** may provide insulation of the conductive pad layer **60** by surrounding the rear edge portion **60**R (corresponding to the thickness of the rear edge portion **60**R) and the rear portion (corresponding to the thickness of the side edge portion **60**S) of the side edge portion **60**S.
(67) In an example embodiment, the step P between the conductive pad layer **60** and the insulating layer **80** may be formed around the uneven pattern **65**.
(68) In another example embodiment, the step P between the conductive pad layer **60** and the insulating layer **80** may be formed around the pressing region rather than around the uneven pattern **65**. That is, because the step P is to provide effective pressurization to the conductive thermocompression bonding layer **50**, formed on the conductive pad layer **60**, by protruding the conductive pad layer **60** relative to the insulating layer **80**, the step P may be formed around the pressing region, which corresponds to a projection region of the pressing tool TO (refer to FIG. **3**), in the conductive pad layer **60**.
(69) For example, the pressing region of the conductive pad layer **60** may correspond to a projection region on which the pressing tool TO is projected in the second direction Z**2** and may correspond to a projection region on which the conductive thermocompression bonding layer **50** to be pressed is projected. In an example embodiment, the pressing region may refer to a portion of the uneven pattern **65** overlapping the conductive thermocompression bonding layer **50**, and may refer to a portion of the uneven pattern **65** overlapping the conductive thermocompression bonding layer **50** in the first direction Z**1**, rather than the entirety of the uneven pattern **65**, and a step P for effective pressurization of the conductive thermocompression bonding layer **50** may be formed around a portion of the uneven pattern **65** overlapping the conductive thermocompression bonding layer **50**. For example, the step P may be formed in a side edge portion **60**S of the conductive pad layer **60** overlapping the conductive thermocompression bonding layer **50** in the third direction Z**3**, rather than being formed in a side edge portion **60**S of the conductive pad layer **60** overlapping the entirety of the uneven pattern **65** in the third direction Z**3**.
(70) The pressing region may extend in the third direction Z**3** in which the first and second electrodes **11** and **12** are arranged. By applying the pressing tool TO (refer to FIG. **3**) along a pressing region extending across the first and second electrodes **11** and **12**, a sufficient pressure may be applied to the conductive thermocompression bonding layer **50** between the first and second electrodes **11** and **12** and the first and second conductive pad layers **61** and **62**. In this case, the pressing tool TO may press the conductive thermocompression bonding layer **50** between the first and second electrodes **11** and **12** and the first and second conductive pad layers **61** and **62** at a time through one pressurization to the pressing region.
(71) In another example embodiment, the pressing tool TO may press the conductive thermocompression bonding layer **50** between the first and second electrodes **11** and **12** and the first and second conductive pad layers **61** and **62** at different times through divided pressurization to the pressing region.
(72) The pressing tool TO may include a heater therein and may provide sufficient heat for connection between the electrode **11** and **12** of the battery cell **10** and the conductive pad layer **60**, in addition to pressurization to the conductive thermocompression bonding layer **50**.
(73) In an example embodiment, the conductive pad layer **60** may be formed on the circuit portion C, and the insulating layer **80** may be formed as a portion of the circuit portion C. Through the

present specification, the conductive pad layer **60** may be formed on the circuit portion C, and even though the insulating layer **80** is formed as a portion of the circuit portion C, the conductive pad layer **60** and the insulating layer **80** may be formed integrally with the circuit portion C. In an example embodiment, the circuit portion C may be formed to have a multilayer structure in which the insulating layer **80** is formed between a plurality of conductive lines, and the conductive pad layer **60** may be formed on the circuit portion C including the insulating layer **80** as one component. For example, a plating layer may be formed on the insulating layer **80** of the circuit portion C, and the conductive pad layer **60** having the uneven pattern **65** formed thereon may be formed by selectively etching the plating layer.

(74) The following comparative example and example are provided in order to highlight characteristics of one or more embodiments, but it will be understood that the comparative example and the example are not to be construed as limiting the scope of the embodiments, nor is the comparative example to be construed as being outside the scope of the embodiments. Further, it will be understood that the embodiments are not limited to the particular details described in the comparative example and the example.

(75) FIGS. **8**A and **8**B illustrate the structures of a comparative example and an example according to an embodiment, for measuring a connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60**, to show technical effects.

(76) Referring to FIGS. **8**A and **8**B, connection resistance values measured on the charge and discharge path of the battery cell **10** were compared by measuring the value of a connection resistance between the electrode **11** of the battery cell **10** and the conductive pad layer **60** on the charge and discharge path of the battery cell **10** from the electrode **11** of the battery cell **10** to the conductive pad layer **60**.

(77) As discussed in detail below, through the comparison of the connection resistance values in the comparative example and the example according to an embodiment, it can be seen that the connection resistance value on the charge and discharge path is reduced in the example, as compared to the comparative example. In this case, the “connection resistance value” may refer to a resistance between the electrode **11** of the battery cell **10** and the conductive pad layer **60**, the resistance being formed by the conductive thermocompression bonding layer **50**.

(78) More specifically, in this experiment, after the self-resistance value of each of the electrode **11** of the battery cell **10** and the conductive pad layer **60** is measured, a connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** is calculated by subtracting the previously measured self-resistance values of the electrode **11** of the battery cell **10** and the conductive pad layer **60** from a resistance value on the entire charge and discharge path between the electrode **11** of the battery cell **10** and the conductive pad layer **60**.

(79) In this experiment, two measurement points EP1 and EP2 were set on the charge and discharge path of the battery cell **10**, which is formed between the electrode **11** of the battery cell **10** and the conductive pad layer **60**. A middle measurement point MP was set at an intermediate position between the electrode **11** of the battery cell **10** and the conductive pad layer **60** where two end measurement points EP1 and EP2 corresponding to end positions on the charge and discharge path were set. The middle measurement point MP corresponds to a connection position between the electrode **11** of the battery cell **10** and the conductive pad layer **60**.

(80) Thus, in detail, three measuring points EP1, EP2, and MP (including two end measurement points EP1 and EP2 and one middle measurement point MP) were set.

(81) A resistance value from the end measurement point EP1 (set on the electrode **11** of the battery cell **10**) to the middle measurement point MP corresponds to the self-resistance value of the electrode **11** of the battery cell **10**.

(82) A resistance value from the end measurement point EP2 (set on the conductive pad layer **60**) to the middle measurement point MP corresponds to the self-resistance value of the conductive pad layer **60**.

(83) A resistance value between the end measurement point EP1 and the end measurement point EP2 corresponds to a resistance value on the entire charge and discharge path.
(84) Based on the above, the resistance value on the entire charge and discharge path refers to a resistance value measured in a connection state between the electrode **11** of the battery cell **10** and the conductive pad layer **60**.
(85) In this experiment, a connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** was calculated by first measuring self-resistance values of the electrode **11** of the battery cell **10** and the conductive pad layer **60**, and then subtracting the previously measured self-resistance values of the electrode **11** of the battery cell **10** and the conductive pad layer **60** from a resistance value on the entire charge and discharge path between the electrode **11** of the battery cell **10** and the conductive pad layer **60**.
(86) In the above, because the connection resistance value between the battery cell **10** and the conductive pad layer **60** is calculated in an indirect method of subtracting the self-resistance values of the electrode **11** of the battery cell **10** and the conductive pad layer **60** from a resistance value on the entire charge and discharge path, rather than a direct measurement method, the connection resistance value may be obtained as a minus (−) value, and the minus (−) value may correspond to an experimental measurement error. That is, as in the experimental results of the disclosure, a minus (−) value may mean that the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** may be reduced to a level close to zero.
(87) Electrode Formed of a Nickel Material:
(88) Tables 1 and 2 below show experimental results of resistance values measured in the comparative example and the example, respectively, for a case in which the electrode **11** of the battery cell **10** was formed of a nickel material.
(89) The number of experiments performed in the comparative example is different from the number performed for the example.
(90) Comparing the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** in the comparative example to the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** in the example, it may be seen that the connection resistance value in the example is remarkably reduced.
(91) That is, in a case in which the conductive thermocompression bonding layer **50** mediating the connection between the electrode **11** of the battery cell **10** and the conductive pad layer **60** is formed on the uneven pattern **65** on the conductive pad layer **60** (as in the example), it may be seen that the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** is significantly reduced, as compared to a case in which the conductive thermocompression bonding layer **50** mediating the connection between the electrode **11** of the battery cell **10** and the conductive pad layer **60** is formed on the base portion **60***a* of the conductive pad layer **60** in which the uneven pattern **65** is not formed (as in the comparative example).
(92) TABLE-US-00001 TABLE 1 resistance on conductive electrode of entire charge and connection No. pad layer battery cell discharge path resistance 1 1.47 5.07 8.67 2.13 2 1.53 4.97 8.17 1.67 3 1.61 5.13 17.65 10.91 (resistance values; unit: mΩ)
(93) TABLE-US-00002 TABLE 2 resistance on conductive electrode of entire charge and connection No. pad layer battery cell discharge path resistance 1 1.89 4.84 6.03 −0.7 2 1.87 5.30 6.24 −0.93 3 1.84 4.58 6.04 −0.38 4 1.90 4.85 6.50 −0.25 5 1.94 4.79 5.83 −0.90 6 1.80 4.83 6.20 −0.43 7 1.97 4.61 6.15 −0.43 8 1.87 4.25 5.62 −0.50 9 1.86 4.47 6.10 −0.23 10 1.77 4.63 6.08 −0.32 (resistance values; unit: mΩ)
Electrode Formed of an Aluminum Material:
(94) Tables 3 and 4 below show experimental results of resistance values measured in the comparative example and the example, respectively, for a case in which the electrode **11** of the battery cell **10** is formed of an aluminum material.
(95) The number of experiments performed in the comparative example is different form that

performed in the embodiment of the disclosure.
(96) Comparing the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** in the comparative example to the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** in the example, it may be seen that the connection resistance value in the embodiment of the disclosure is remarkably reduced.
(97) That is, in a case in which the conductive thermocompression bonding layer **50** is formed on the uneven pattern **65** on the conductive pad layer **60** (as in the example), it may be seen that the connection resistance value between the electrode **11** of the battery cell **10** and the conductive pad layer **60** is significantly reduced, as compared to a case in which the conductive thermocompression bonding layer **50** is formed on the base portion **60***a* of the conductive pad layer **60** in which the uneven pattern **65** is not formed (as in the comparative example).
(98) TABLE-US-00003 TABLE 3 resistance on conductive electrode of entire charge and connection No. pad layer battery cell discharge path resistance 4 1.51 1.63 8.22 5.08 5 1.57 1.58 89.20 86.05 6 1.52 1.44 9.80 6.84 (resistance values; unit: mΩ)
(99) TABLE-US-00004 TABLE 4 resistance on conductive electrode of entire charge and connection No. pad layer battery cell discharge path resistance 11 2.06 1.69 2.93 −0.82 12 2.06 1.63 2.82 −0.87 13 1.99 1.49 2.89 −0.59 14 2.03 1.49 2.92 −0.60 15 2.10 1.60 2.91 −0.79 16 2.11 1.40 2.71 −0.80 17 1.93 1.42 2.90 −0.45 18 1.94 1.53 2.89 −0.58 19 1.95 1.55 2.85 −0.65 20 2.03 1.48 2.85 −0.66 (resistance values; unit: mΩ)
(100) As described above, embodiments may provide a battery pack in which a connection process may be easily performed while increasing reliability of an electrical connection between a battery cell and a circuit portion forming a charge and discharge path of the battery cell.
(101) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A battery pack, comprising: a battery cell including an electrode extending in a first direction; a circuit portion connected to the electrode; and a connection portion connecting between the electrode and the circuit portion, the connection portion including: a conductive thermocompression bonding layer, and a conductive pad layer having a concave accommodation space in a second direction intersecting with the first direction to accommodate a first portion of the conductive thermocompression bonding layer, at least a second portion of the conductive thermocompression bonding layer being directly between the electrode and the conductive pad layer, wherein the conductive pad layer includes convex portions relatively protruding in the second direction and concave portions relatively concave in the second direction, the convex and concave portions are arranged in an alternating pattern and are directly connected to each other, the convex and concave portions are integral with each other and include a same metal, and the conductive thermocompression bonding layer includes conductive particles supported on the convex portions and an insulating resin accommodated in the concave portions.
2. The battery pack as claimed in claim 1, wherein: the electrode includes a first electrode and a second electrode, arranged in a third direction intersecting with the first and second directions, the

first and second electrodes having different polarities, and the conductive pad layer includes a first conductive pad layer and a second conductive pad layer, which are apart from each other in the third direction so as to be connected to the first electrode and the second electrode, respectively.
3. The battery pack as claimed in claim 2, wherein the conductive thermocompression bonding layer is continuously formed in the third direction between the first electrode and the first conductive pad layer and between the second electrode and the second conductive pad layer.
4. The battery pack as claimed in claim 1, wherein the concave portions provide the concave accommodation space.
5. The battery pack as claimed in claim 1, wherein longitudinal directions of the convex portions and the concave portions extend in parallel to each other in the first direction and in parallel to an extension direction of the electrode.
6. The battery pack as claimed in claim 1, wherein the conductive pad layer includes an uneven pattern in which the convex portions and the concave portions are alternately arranged in a third direction intersecting with the first and second directions, the convex and concave portions alternating in a direction parallel to a short side of the battery cell through which the electrode extends.
7. The battery pack as claimed in claim 6, wherein the uneven pattern is formed at a position biased close to a front position of the conductive pad layer facing the battery cell.
8. The battery pack as claimed in claim 6, wherein the uneven pattern is completely surrounded by an edge portion of the conductive pad layer, a top surface of the edge portion of the conductive pad layer being coplanar with top surfaces of the convex portions.
9. The battery pack as claimed in claim 1, wherein an insulating layer is formed around the conductive pad layer.
10. The battery pack as claimed in claim 9, wherein at least some of edge portions of the conductive pad layer forming a boundary between the conductive pad layer and the insulating layer form a step so that an upper surface of the conductive pad layer protrudes more prominently than an upper surface of the insulating layer in the second direction.
11. The battery pack as claimed in claim 10, wherein the step is formed around an uneven pattern formed on the conductive pad layer.
12. The battery pack as claimed in claim 10, wherein the step is formed around the conductive thermocompression bonding layer.
13. The battery pack as claimed in claim 10, wherein the edge portions of the conductive pad layer include: a front edge portion formed at a position relatively close to the battery cell in the first direction; a rear edge portion formed at a position relatively far from the battery cell in the first direction; and a side edge portion connecting the front edge portion to the rear edge portion.
14. The battery pack as claimed in claim 13, wherein: the step is formed around an uneven pattern formed on the conductive pad layer, and the step is formed in a front portion of the side edge portion overlapping the uneven pattern in a third direction intersecting with the first and second directions.
15. The battery pack as claimed in claim 13, wherein the step is formed in a front portion of the side edge portion overlapping the conductive thermocompression bonding layer in a third direction intersecting with the first and second directions.
16. The battery pack as claimed in claim 13, wherein the step is formed in the front edge portion relatively close to the battery cell in the first direction.
17. The battery pack as claimed in claim 1, wherein: the conductive pad layer is between the circuit portion and an edge of the electrode, the edge of the electrode extending outside the battery cell, and the conductive pad layer includes a first surface and a second surface opposite each other, the first surface including an uneven pattern with alternating convex and concave portions facing the edge of the electrode, and the second surface being entirely flat and facing the circuit portion.